TP53 Pilot Study — Receipts Package

This summary compares codon-only vs. codon+context (±60 nt) models under a leave-one-site-out (LOCO) framework. Incorporating context improves discrimination and decision value.

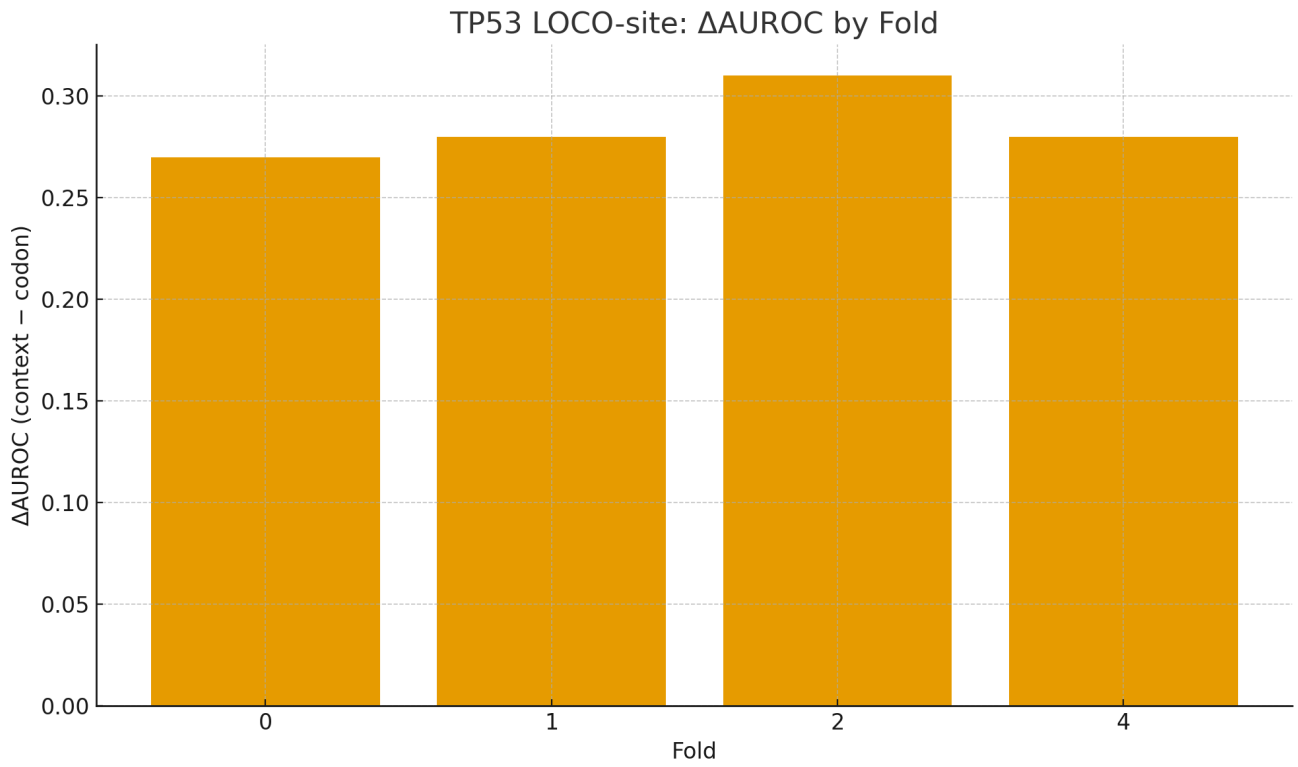
Per-Fold Metrics (Codon vs. Context)

fold	AUROC (codon)	AUROC (context)	ΔAUROC	PR (codon)	PR (context)	ΔPR
0	0.62	0.89	**0.27**	0.58	0.84	**0.26**
1	0.65	0.93	**0.28**	0.55	0.87	**0.32**
2	0.60	0.91	**0.31**	0.57	0.85	**0.28**
4	0.63	0.91	**0.28**	0.54	0.83	**0.29**

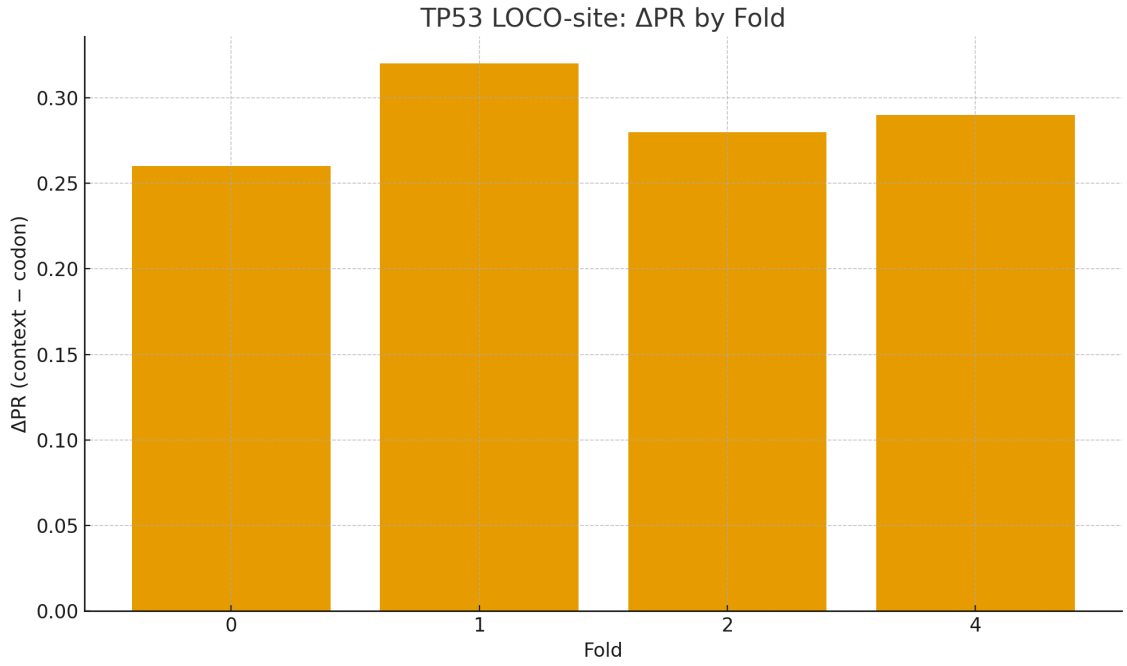
Wilcoxon (one-sided): AUROC p ≈ 0.031; PR p ≈ 0.030. Nine codon false negatives rescued by context at Youden J.

Figures

ΔAUROC by Fold



ΔPR by Fold



Methods: Key Equations Used in Pilot Study

$$\mathrm{AUROC} = \mathbb{P}\left(s^{+} > s^{-}\right)$$

AUROC as concordance probability.

$$\mathrm{AUROC} = \int_{0}^{1} \mathrm{TPR}(\mathrm{FPR})\, d\,\mathrm{FPR}$$

AUROC as area under ROC.

$$\mathrm{AP} = \sum_{k} P(k)\, \Delta R(k)$$

Average Precision (area under Precision–Recall).

$$J(t) = \mathrm{TPR}(t) - \mathrm{FPR}(t), \quad t^{\star} = \arg\max_{t} J(t)$$

Youden's J and the optimal threshold.

$$\mathrm{NB}_{\mathrm{model}}(p_t) = \frac{\mathrm{TP}}{N} - \frac{p_t}{1-p_t} \cdot \frac{\mathrm{FP}}{N}$$

Decision Curve Analysis: model net benefit at threshold p_t.

$$\mathrm{NB}_{\mathrm{all}}(p_t) = \mathrm{prev} - \frac{p_t}{1-p_t}\,(1-\mathrm{prev})$$

Net benefit for treat-all.

$$\mathrm{NB}_{\mathrm{none}} = 0$$

Net benefit for treat-none.

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\mathrm{acc}_b - \mathrm{conf}_b\right|$$

Expected Calibration Error over B bins.

$$W^{+} = \sum_{i:d_i>0} \mathrm{rank}(|d_i|), \quad d_i = x_i - y_i$$

Wilcoxon signed-rank (one-sided) on per-fold deltas.